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Which is the better forecasting model? A comparison between HAR-RV and multifractality volatility

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HIGHLIGHTS

- Compared the forecasting performance of the multifractal volatility and HAR-RV models.
- Used the MCS test to explore the forecasting ability of the high-frequency volatility models.
- The HAR-RV models have greater forecasting accuracy than multifractal volatility models.

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ABSTRACT

In this paper, by taking the 5-min high frequency data of the Shanghai Composite Index as example, we compare the forecasting performance of HAR-RV and Multifractal volatility, Realized volatility, Realized Bipower Variation and their corresponding short memory model with rolling windows forecasting method and the Model Confidence Set which is proved superior to SPA test. The empirical results show that, for six loss functions, HAR-RV outperforms other models. Moreover, to make the conclusions more precise and robust, we use the MCS test to compare the performance of their logarithms form models, and find that the HAR-log(RV) has a better performance in predicting future volatility. Furthermore, by comparing the two models of HAR-RV and HAR-log(RV), we conclude that, in terms of performance forecasting, the HAR-log(RV) model is the best model among models we have discussed in this paper.

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1. Introduction

Modeling and forecasting volatility in financial markets play a central role in many important fields, such as asset pricing, asset allocation and risk measurement. Fundamentally speaking, since the real latent volatility of the market is unobservable, the characterization and prediction of volatility are hot topics and poses as challenges in modern financial research. In these works, Engle's autoregressive conditional heteroskedasticity model (ARCH) [1], Bollerslev's generalized autoregressive conditional heteroskedasticity model (GARCH) [2] as well as Taylor's stochastic volatility models (Stochastic volatility model, SV) [3], in early days, were the dominant approaches to model the volatility. Even now, the three models are still being widely employed in forecasting volatility in related field.

To better describe the stylized facts of volatility, scholars studied much about GARCH and SV models, and put forward corresponding expansion models, such as EGARCH [4], GJR-GARCH [5], APARCH [6], FIAPARCH [7], HYGARCH [8] and so forth. It is imperative to point out that both the early GARCH and SV models and the extended models are based on low-frequency data (especially, daily data). They used these data to characterize and forecast future conditional volatility. However, this resulted in a great loss of data on inter-day trading information and also caused bias in estimating and forecasting the conditional volatility.

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Over the past decade, the technique of constructing volatility models based on high-frequency financial data has been drawn wide attention of many scholars. Among these techniques, Realized Volatility (RV), first proposed by Andersen et al. [9,10], and defined as the sum of all available intraday high frequency squared returns. It is widely accepted that, under suitable conditions (for example, no microstructure noise and jump), realized volatility is an unbiased and highly efficient estimator of integrated variance, as discussed in Ref. [10] as well as in concurrent work by Barndorff-Nielsen and Shephard [11]. In fact, Realized Volatility was primarily used as a proxy variable of the actual volatility to evaluate forecasting ability of models (such as GARCH model, etc.). While with the availability of high-frequency data and the advantages of Realized Volatility, Ref. [10,12,13] built the so-called dynamic models (ARFIMA-RV) based on Realized Volatility. And compared with low-frequency data volatility models such as GARCH and SV, studies indicated that ARFIMA-RV was superior in predication.

However, Corsi [14] pointed out that the fractional integrated model (like ARFIMA) was just a convenient mathematical trick and a clear economic interpretation was absent. Moreover, the fractional integrated model needed to build a fractional difference filter, which would lead to losses of a large number of observations or transaction information. It was worth mentioning that Corsi [14], based on Heterogeneous Market Hypothesis, proposed a simple Heterogeneous Autoregressive model of Realized Volatility (HAR-RV). To compare with ARFIMA model, the HAR-RV model not only avoided the complex estimation of ARFIMA model, but also successfully reproduced the typical characteristics of financial return series, such as long memory and fat tail. In addition, by using the autoregressive processes of realized volatility on three volatility components (the daily, weekly and monthly), the economic interpretation is that each volatility component in the cascade responsive to a market component that forms expectations for the next period's volatility based on the observation of the current realized volatility and on the expectation for the longer horizon volatility. Since its proposal, scholars have further studied and applied the HAR-RV model. For example, Andersen et al. [15] broke down the realized volatility into continuous and jump components, employed the HAR-RV-CJ model to predict the realized volatility and concluded that the predictability of volatility derived mainly from the continuous component. McAleer and Medeiros [16] expanded the HAR-RV model, advancing with a multiple regime smooth transition heterogeneous autoregressive model to capture the non-linear and long-term autocorrelation features of time series.

In the financial market, because most scholars had found significant and complex fractal characteristics in the market, fractal was considered to be a "stylized fact" [17–21]. Interestingly, fractal exists not only in mature stock markets but also in emerging stock markets, but the fractal strength is weaker in mature markets than in emerging markets [22–26]. One thing worth mentioning is that all the Chinese market fractal studies drew almost the same conclusion: both Shanghai and Shenzhen stock market were significantly featured by fractal feature. As early as in the 1997, Mandelbrot, "father of the fractal theory", forecasted the great prospects in the financial sector in his book *Fractals and Scaling in Finance* [27]. Later on, Mandelbrot [28] wrote in *Scientific American*, a globally leading scientific journal, illustrating that the fractal theory was a powerful tool to quantitatively characterize various complicated volatile characteristics in financial markets. Given the fact that multifractal is a powerful tool for depicting volatility complexities in financial markets, Wei and Wang [29] proposed a measurement method and used the ARFIMA models to model multifractal volatility (MFV) based on this theory. It was evidenced that the forecasting ability of MFV models was significantly better than GARCH and SV models. Besides, the measurement of the MFV and its dynamic model was applied in risk management [30], and other areas.

It is important to emphasize that even though both HAR-RV and ARFIMA-MFV had good forecasting performance, no comparisons or contrast analysis of these two models were ever conducted. In other words, no one ever questioned which model had a better performance. This paper first compared the forecasting ability of the HAR-RV and ARFIMA-MFV model, aiming to find a proper volatility model to describe and predict the volatility of the Chinese stock market because elaborating the market volatility was extremely important in risk measurement and asset pricing, and other areas. Furthermore, Ashely [31], Lo [32] and Foster [33] found that the testing of in-sample goodness-of-fit was usually affected by data mining bias but the out-of-sample rolling-forecasting method could refrain from this error. Therefore, this paper utilized in-sample estimation and out-of-sample rolling windows forecasting method to evaluate the forecasting ability of HAR-RV, ARFIMA-MFV and other high-frequency volatility models (such as ARFIMA-RV). Meanwhile, the Model Confidence Set (MCS) method was also adopted, proposed by Hansen et al. [34], to compare the forecasting accuracy of these models. Compared with the Superior Predictive Ability (SPA), which also was proposed by Hansen and Lunde [35], the MCS test had looser conditions and there was no need to choose the base model, therefore the result of MCS test is more robust and rigorous. This paper adopted the MCS method to ensure the robustness of the conclusions as well as to enrich the testing methods for volatility prediction ability.

This paper is organized as follows. Section 2 illustrates the volatility measurement methods and its econometric models. Section 3 interprets the MCS testing method. Section 4 outlines the empirical analysis. The last section presents the conclusion.

2. Volatility measure and its model

2.1. Realized volatility and the ARFIMA model

The fact that the measures of volatility are latent makes it difficult to evaluate the performance of volatility models. A common solution to this problem is to substitute a proxy for the true volatility and evaluate the models by comparing their

predicted volatility to the proxy. However, it is generally accepted that squared daily returns provide a poor approximation of actual daily volatility. Furthermore, Andersen and Bollerslev [36] pointed out that more accurate estimates can be obtained by summing all squared intraday returns. If we were to apply their method directly in this paper, then we would define the RV measurement as

$$R_t = 100(\ln P_t - \ln P_{t-1}) \tag{1}$$

$$R_{t,d} = 100(\ln P_{t,d} - \ln P_{t,d-1}), \quad d = 1, 2, \ldots, 48 \tag{2}$$

$$\mathrm{RV}'_t = \sum R^2_{t,d}. \tag{3}$$

However, this definition neglects the information about overnight returns. To address this problem, Hansen and Lunde [37] suggested scaling the RV measurement in the following way.

$$\mathrm{RV}_t = \gamma \mathrm{RV}'_t, \tag{4}$$

where the so-called scale parameter γ is defined as

$$\gamma = \frac{N^{-1}\sum_{t=1}^{N} R_t^2}{N^{-1}\sum_{t=1}^{N} \mathrm{RV}'_t}. \tag{5}$$

Based on the value of AIC, SC and maximum likelihood, we use the ARFIMA(1, d, 1) to model realized volatility. The ARFIMA(1, d, 1) model with mean μ for realized volatility can be given by

$$\varphi(L)(1-L)^d(Y-u) = \Theta(L)\varepsilon_t \tag{6}$$

$$\varphi(L) = 1 - \varphi_1 L - \varphi_2 L^2 - \cdots - \varphi_p L^p \tag{7}$$

$$\Theta(L) = 1 + \theta_1 L + \theta_2 L^2 + \cdots + \theta^q \tag{8}$$

where L is the lag operator, coefficients d, φ and θ are fixed and unknown and ε_t is Gaussian white noise with mean zero and variance σ_ε^2. Moreover, Andersen et al. [9] found the distribution of RV to be highly non-normal and skewed, but its logarithms to be approximately normal. Accordingly, they suggested that the natural logarithms of a RV measurement series, denoted as log(RV), could be modeled by a Gaussian dynamic process. They adopted an ARFIMA to model this process. In this paper, we not only research the model ARFIMA-RV, but also consider the ARFIMA-log(RV). Furthermore, to ensure the robustness of the conclusions, we also introduce the ARMA-RV and ARMA-log(RV) to this paper.

Moreover, in recent years, the literature on volatility modeling has put great emphasis on the wide availability of intra-daily financial returns data, so the realized variance achieves a great development and improvement. Especially, Barndorff-Nielsen et al. [38,39] proposed realized bipower variation (RBV), which has all advantages of realized volatility, and also has a better robustness than RV. In this paper, we use the ARFIMA(1, d, 1) and ARMA to model realized bipower variation (and their logarithmic form), ARFIMA-RBV and ARMA-RBV (ARFIMA-log(RBV) and ARMA-log(RBV)).

2.2. Heterogeneous autoregressive model of realized volatility

Corsi [14] proposed Heterogeneous Autoregressive model of Realized Volatility (HAR-RV) based on Heterogeneous Market Hypothesis that is analyzed by Müller et al. [40,41]. The HAR-RV model leads to a simple AR-type model in the realized volatility with the feature of considering volatilities realized over different time horizons, which are one day (1 day), one week (5 days) and one month (22 days). To simplify, the HAR-RV model is made up of three heterogeneous volatility components. The HAR-RV model is,

$$\mathrm{RV}^d_{t+1} = c + \beta^d \mathrm{RV}^d_t + \beta^w \mathrm{RV}^w_t + \beta^w \mathrm{RV}^m_t + \omega_t. \tag{9}$$

Moreover, the RV^w_t and RV^m_t are calculated as below

$$\mathrm{RV}^w_t = \frac{1}{5}(\mathrm{RV}^d_{t-1} + \mathrm{RV}^d_{t-2} + \mathrm{RV}^d_{t-3} + \mathrm{RV}^d_{t-4} + \mathrm{RV}^d_{t-5}) \tag{10}$$

$$\mathrm{RV}^m_t = \frac{1}{22}(\mathrm{RV}^d_{t-1} + \mathrm{RV}^d_{t-2} + \cdots + \mathrm{RV}^d_{t-3} + \mathrm{RV}^d_{t-4} + \mathrm{RV}^d_{t-22}). \tag{11}$$

The HAR-RV model seems to successfully achieve the purpose of modeling the long memory behavior of volatility in a very simple and parsimonious way. Moreover, we can get the form of logarithmic HAR-RV model,

$$\log(\mathrm{RV}^d_{t+1}) = c + \beta^d \log(\mathrm{RV}^d_t) + \beta^w \log(\mathrm{RV}^w_t) + \beta^w \log(\mathrm{RV}^m_t) + \omega_t. \tag{12}$$

2.3. Multifractal volatility measure and its model

Recent studies discovered that the multifractal spectrum of high-frequency price fluctuations over the course of a trading day contains valuable volatility information [42,43]. Thus, in this section, we discuss how to construct a multifractal volatility

measure (MFV) from a multifractal spectrum and how to model the MFV series according to the method by Wei and Wang [29].

In contrast with the realized volatility measurement proposed by Andersen et al. [9], the MFV measure is constructed on the basis of high-frequency price data as opposed to high-frequency return data. Following the steps in Ref. [29], we first compute the multifractal spectrum for one day using high-frequency price data within the day before calculating MFV from the multifractal spectrum.

We use the box-counting method proposed in Ref. [42] to calculate the multifractal spectrum of the SSEC. To be clear, we denote the high-frequency price quotes for the SSEC at time t (at 5-min intervals) as $I(t)$ and t running from 1 to $M = 48 \times 3164 = 151\,872$. The price variation within a trading day can be divided into many normalized boxes (time intervals) of size δ ($\delta \leqslant 1$). For example, in this case the box size can be 1/48, 1/24, 1/16, 1/12, 1/8, 1/6, 1/4, 1/3, 1/2, or 1. Suppose we need m boxes to cover the 48 high-frequency quotes within a trading day $I(t)$, $t = 1, 2, \ldots, 48$ and there are n quotes in each box. If P_i is the average probability in box i, then

$$P_i(\delta) = \sum_{j=1}^{n} I(i_j) \Big/ \sum_{t=1}^{48} I(t), \quad i = 1, 2, \ldots, m \tag{13}$$

where $I(i_j)$ is the jth quote in the ith box in the trading day.

We can then describe the average probability in box i from a multifractal perspective as

$$P_i(\delta) \sim \delta^{\alpha} \tag{14}$$

$$N_{\alpha}(\delta) \sim \delta^{-f(\alpha)}. \tag{15}$$

According to Wei and Wang's suggestion [29], what is important for us is statistical information about index fluctuations contained in multifractal spectra $f(\alpha)$. We can use the partition function $S_q(\delta)$ to calculate such statistics. The partition function $S_q(\delta)$ is defined and expressed as a power law of δ with an exponent $\tau(q)$, where q is the moment order

$$S_q(\delta) = \sum_{i=1}^{m} P_i^q(\delta) \sim \delta^{\tau(q)} \tag{16}$$

and $\tau(q)$ can be obtained from the slope of the linear part of the $\ln S_q(\delta) - \ln \delta$ curve. Then $f(\alpha)$ can be obtained by performing a Legendre transformation as follows:

$$\alpha = \mathrm{d}\tau(q)/\mathrm{d}q \tag{17}$$

$$f(\alpha) = \alpha q - \tau(q). \tag{18}$$

Therefore the width of the multifractal spectrum can be expressed as $\Delta\alpha = \alpha_{\max} - \alpha_{\min}$, where $\alpha_{\max}$ and $\alpha_{\min}$ are the maximum and minimum of α. Recently Jiang and Zhou [44,45] argued that if the width of the singularity is very narrow, it means that there is actually no singularity and thus no multifractality. Furthermore, they indicated that there is no multifractality when the price data are adopted and the correct way is to use the absolute returns in applying the partition function approach. More detailed analysis on the sources of multifractality of financial markets can be found in Refs. [46–48]. However we do not carry out the following analysis along with the routine adopted in the above papers [44–48]. The main purpose of this manuscript is not to discuss the source of multifractality in financial markets or the way to apply the partition function approach. We focus on the statistical information obtained in the multifractal spectrum of intraday high-frequency price data. Even if the width of the singularity is very narrow, we can also extract useful information from it.

Since the probability of each box is $P_i(\delta) \sim \delta^{\alpha}$ and $\delta < 1$, $\alpha_{\max}$ and $\alpha_{\min}$ denote the values of the maximum and minimum probabilities. In other words, $\alpha_{\min}$ indicates the highest "price level" in that trading day and $\alpha_{\max}$ indicates the lowest "price level" in that trading day. Thus the larger the value of $\Delta\alpha$, the more violent the price fluctuation in the trading day, and $\Delta\alpha$ can be treated as a measure of daily volatility.

To make $\Delta\alpha$ comparable to realized volatility and other historical volatility measures, similar to the means of scaling realized volatility in Section 2.1, we formally define multifractal volatility measure (MFV) for day t as

$$\mathrm{MFV}_t = \lambda_1 \Delta_{\alpha_t} \tag{19}$$

$$\lambda_1 = \frac{N^{-1} \sum_{t=1}^{N} R_t^2}{N^{-1} \sum_{t=1}^{N} \Delta_{\alpha_t}}. \tag{20}$$

According to Wei and Wang [29], we use a widely adopted autoregressive fractionally integrated moving average model (ARFIMA) that has been shown to be effective in time series modeling [9,12] to describe the dynamics of time series MFV. The form of the ARFIMA-MFV model looks like Eq. (6). Moreover, we construct the ARFIMA-log(MFV) model to explore the performance forecasting. Furthermore, we also consider the ARMA short memory model to describe the multifractal volatility (ARMA-MFV and ARMA-log(MFV)).

Andersen and Bollerslev [36] and Barndorff-Nielsen and Shephard [49] demonstrated that as the proxy of actual volatility, realized volatility has less noise than daily squared returns. Thus, it may be more reasonable to construct a volatility measure

by rescaling the degrees of multifractality using the expectation of RV, so Chen and Wu [50] proposed a new multifractal volatility (MV), which is different from Wei and Wang [29], the definition goes as follows:

$$\mathrm{MV}_t = \frac{\Delta_{\alpha_t}}{E(\Delta_{\alpha_t})} E(\mathrm{RV}_t). \tag{21}$$

In Eq. (21), $\mathrm{E}(\mathrm{RV}_t)$ and $\mathrm{E}(\Delta\alpha_t)$ denote the average of the two whole time series RV_t and $\Delta\alpha_t$. We use the ARFIMA and ARMA to model the MV (ARFIMA-MV and ARMA-MV). Moreover, we also consider those models' logarithms form, ARFIMA-log(MV) and ARMA-log(MV).

As we all know, neglecting microstructure noise in calculating realized volatility (RV) can lead to biased and inconsistent estimates of integrated volatility (IV) as a true measure of daily volatility. Andersen and Bollerslev [36] have showed that ex post daily volatility is best measured by aggregating 288 squared 5 min returns. The five-minute frequency was suggested as a trade-off between accuracy, which is theoretically optimized using the highest possible frequency, and microstructure noise that can arise through the bid–ask bounce, asynchronous trading, infrequent trading, and price discreteness, among other factors. To reduce the effect of microstructure noise, we choose the five-minute frequency as our research object in this paper.

3. Forecasting methodology and the MCS test

In this section, we evaluate the forecasting performance of the HAR-RV, ARFIMA-RV, ARFIMA-MFV, ARFIMA-MV and their ARMA-type models. Those models all are based on intraday high-frequency data (5-min frequency). Following is the forecasting process.

The January 1, 2000–March 31, 2013 observations of Chinese stock market are classified into two subgroups: (1) in-sample data for volatility modeling, covering 2800 trading days; (2) out-of-sample data for model evaluation, covering the last 364 trading days of the total data sample. The estimation period is then rolled forward by adding one new day and dropping the most distant day. In this way, the sample size used to estimate the models remains at a fixed length and the forecasts do not overlap, thereby allowing daily (one-day-ahead) out-of-sample volatility forecasts to be obtained.

The volatility forecasts obtained by the RV, MFV, MV and ARMA-type models are indicated by $\hat{\sigma}_t^2$ (one-day ahead volatility forecasts), and the scaled RV measurement denoted as RV_t is taken as a proxy for actual daily volatility (the forecasting benchmark). Various forecasting criteria or loss functions can be considered in assessing the predictive accuracy of a volatility model, although, as Lopez [51] noted, it is not obvious which loss function is most appropriate for the evaluation of such models. Rather than making a single choice, we thus employ the following six accuracy statistics or loss functions as our forecasting criteria.

$$L_1 : \mathrm{MSE} = M^{-1} \sum_{m=H+1}^{M} (\mathrm{RV}_m - \hat{\sigma}_m^2)^2 \tag{22}$$

$$L_2 : \mathrm{MAE} = M^{-1} \sum_{m=H+1}^{H+M} |\mathrm{RV}m^- \hat{\sigma}_m^2| \tag{23}$$

$$L_3 : \mathrm{HMSE} = M^{-1} \sum_{m=H+1}^{H+M} (1 - \hat{\sigma}_m^2/\mathrm{RV}_m)^2 \tag{24}$$

$$L_4 : \mathrm{HMAE} = M^{-1} \sum_{m=H+1}^{H+M} |1 - \hat{\sigma}_m^2/\mathrm{RV}_m| \tag{25}$$

$$L_5 : \mathrm{QLIKE} = M^{-1} \sum_{m=H+1}^{H+M} (\ln(\hat{\sigma}_m^2) + \mathrm{RV}_m/\hat{\sigma}_m^2) \tag{26}$$

$$L_6 : R^2LOG = M^{-1} \sum_{m=H+1}^{H+M} (\ln(\mathrm{RV}_m)/\hat{\sigma}_m^2)^2 \tag{27}$$

where n is the number of forecasting data points; MSE and MAE are the mean square error and mean absolute error; HMSE and HMAE are the MSE and MAE, respectively, adjusted for heteroskedasticity. Different criteria serve for different practical purposes. For example, in the case of Value-at-Risk applications, greater interest may lie in the accurate forecasting of a high rather than a low level of volatility, which implies that the MSE criterion is the most relevant loss function in risk management applications. Additional discussion of these criteria can be found in Bollerslev et al. [52].

When a particular loss function is smaller for model A than it is for model B, it can be concluded that the forecasting performance of the former is superior to that of the latter. Such a conclusion cannot be made on the basis of a single loss function and a single sample. Recent work has focused on a testing framework that can determine whether one particular

model is outperformed by another [53–55]. As discussed in the Introduction, The Superior Predictive Ability (SPA) test, an extension of the White framework proposed by Hansen and Lunde [37], has been shown to possess good power properties and to be more robust than previous approaches.

However, SPA test is designed to address whether a particular benchmark is significantly outperformed by any of the alternatives used in the comparison. Recently, Hansen et al. [34] proposed a new test method to compare the volatility models, which named Model Confidence Set (MCS). Compared with SPA test, the MCS test has many attractive features. First, MCS test does not require a benchmark to be specified, which is very useful in applications without an obvious benchmark. Second, MCS test acknowledges the limitations of the data. Third, the MCS procedure is that it allows for the possibility that more than one model can be the "best". To save space, this paper includes no further technical details of the SPA test, but more in-depth discussions can be found in the studies of Ref. [34]. In this paper, we take the MCS test in comparing volatility models to robust our conclusions.

Consider a set, M_0, that contains a finite number of models, indexed by $i = 1, \ldots, m_0$. The objects are evaluated over the sample $t = 1, \ldots n$, in terms of a loss function and we denote the loss that is associated with object i in period t as $L_{i,t}$. Define the relative performance variables $d_{ij,t} \equiv L_{i,t} - L_{j,t,t}$ for all $i, j \in M_0$. Then the set of superior objects is defined by

$$M^* \equiv \{u \in M_0 : E(d_{i,uv,m}) \leq 0 \text{ for all } v \in M_0\}. \tag{28}$$

This is done through a sequence of significance tests, where objects that are found to be significantly inferior to other elements of M_0 are eliminated. The hypotheses that are being tested take the form,

$$H_{0,M} : E(d_{i.uv,m}) = 0 \quad \text{for all } u, v \in M \subset M_0. \tag{29}$$

The MCS procedure is based on an equivalence test δ_M and elimination rule e_M. The equivalence test, δ_M, is used to test the hypothesis H_0, M for any $M \subset M_0$, and e_M identifies the object of M that is to be removed from M, in the event that H_0, M is rejected. Hence, the MCS Algorithm is based on the following three steps:

Step 1: Initially set $M = M_0$.

Step 2: Test H_0, M using δ_M at level α.

Step 3: If H_0, M is "accepted" we define the $\hat{M}^*_{1-\alpha} = M$, otherwise we use e_M to eliminate an object from M and repeat the procedure beginning with Step 2.

The set, $\hat{M}^*_{1-\alpha}$, which consists of the set of "surviving" objects (those that survived all tests without being eliminated) is referred to as the model confidence set. In Ref. [34], the significant level α is equal to 0.1. And if p-value is larger than 0.1, the corresponding model is "surviving" model, which means that the performance of forecasting is good. The higher the p-value, the better the forecasting ability of the model.

Moreover, we use the Range Statistic and Semi-quadratic Statistic, defined that,

$$T_{R=} \max_{u,v \in M} \frac{|\bar{d}_{i,uv}|}{\sqrt{\text{var}(d_{i,uv})}}, \qquad T_{SQ} = \max_{u,v \in M} \frac{(\bar{d}_{i,uv})^2}{\text{var}(\bar{d}_{i,uv})}, \qquad \bar{d}_{i,uv} = \frac{1}{M} \sum_{m=H+1}^{H+M} d_{i,uv,m}. \tag{30}$$

The p-value of T_R and T_{SQ} is larger than the 0.1, we can get that the null hypothesis (Eq. (29)) cannot be rejected. The asymptotic distributions of the test statistics, T_R and T_{SQ}, are non-standard because they depend on nuisance parameters (under both the null and the alternative). However, this poses no obstacle as their distributions are easily estimated using bootstrap methods that implicitly solve the nuisance parameter problem. For details about the implementation of the bootstrap we refer to Ref. [34].

4. Empirical results

4.1. Data description and MF-DFA

This paper takes the every 5-min high frequency data of the Shanghai Stock Exchange Composite Index (SSEC) as an example. The data for our empirical study comes from the China Centre for Economic Research (CCER)-high frequency database of stock market, during the period from 1 January 2000 to 31 March 2013, which contains totally $N = 3164$ trading days. The Shanghai Stock Exchange opens from 9:30 to 11:30 am and then from 1:00 to 3:00 pm, so there are 4 trading hours in a trading day, and 48 quotes (per 5 min) of index in a business day (excluding the open price). And there are 151 872 high frequency quotes of the total samples.

Table 1 shows each volatility series' descriptive statistics of this paper. Table 1 demonstrates that all series have the significant feature of skewed, leptokurtic and auto-correlations. So we can get that realized volatility and different multifractal volatility and their logarithm series all contain significant long memory characteristics. Moreover, we also find that all volatility after logarithmic treatment is closer to normal distribution. Furthermore, the results of augmented Dickey–Fuller and Phillips–Perron unit root tests reject the null hypothesis of a unit root in these series. Furthermore, under the significant level ($\alpha = 1\%$), the results of the KPSS test show that the series are stationary, indicating that they are stationary and may be modeled directly without further transformation.

Moreover, we use the multifractal detrended fluctuation analysis (MF-DFA) to explore the multifractality property of stock market returns. And more details about the MF-DFA method see in Refs. [56,57].

Table 1
Descriptive statistics of every volatility series.

	Mean	St. dev.	Skewness	Kurtosis	J–B	ADF	P–P	KPSS	$Q(5)$	$Q(10)$	$Q(20)$
RV	2.649	4.964	6.063*	57.876*	460991*	−6.610*	−61.434*	0.5252	774.0*	1189.7*	1964.9*
log(RV)	0.165	1.230	0.250*	−0.097	34.240*	−5.4317	−65.651*	0.7091	2499.3*	4516.9*	7653.9*
RBV	2.646	4.989	6.317*	63.366*	550393*	−6.512*	−61.767*	0.5297	768.0*	1176.6*	1966.8*
log(RBV)	0.169	1.228	0.238*	−0.078	30.579*	−5.403*	−65.731*	0.7174	2526.4*	4588.5*	7793.7*
RV^w	2.655	3.135	3.245*	14.608*	33686*	−5.618*	−13.604*	0.5350	8076.7*	10798.2*	15501.9*
$\log(RV^w)$	0.517	0.941	0.189	−0.151*	21.917*	−5.857*	−12.455*	0.7394	9291.1	14430.3	22444.1
RV^m	2.658	2.377	1.930*	3.568*	3644.2*	−3.524*	−5.313*	0.5489	14680.4*	26864.8*	43631.6*
$\log(RV^m)$	0.659	0.789	0.172*	−0.274*	25.666*	−3.906*	−5.099*	0.6889	14724.8*	27233.6*	45042.4*
MVF	2.645	3.967	4.098*	26.402*	100750*	−6.285*	−63.138*	0.5247	1156.3*	1855.3*	3089.9*
log(MVF)	0.275	1.176	0.152*	−0.310*	24.629*	−5.355*	−65.287*	0.7062	2647.0*	4752.6*	8007.9*
MV	2.646	3.969	4.098*	26.402*	100750*	−6.285*	−63.138*	0.5247	1156.3*	1855.3*	3089.9*
log(MV)	0.275	1.197	0.152*	−0.314	25.185*	−5.428*	−65.287*	0.7062	2647.0*	4752.6*	8007.9*

The Jarque–Bera statistic tests for the null hypothesis of normality for the distribution of the series. $Q(n)$ is the Ljung–Box statistic for up to n-th order serial correlation. ADF and P–P are statistics for the Augmented Dickey–Fuller and Phillips–Perron unit root tests based on the least AIC criterion, respectively.
* Denote significance at 1% level. KPSS has the null hypothesis of stationarity.

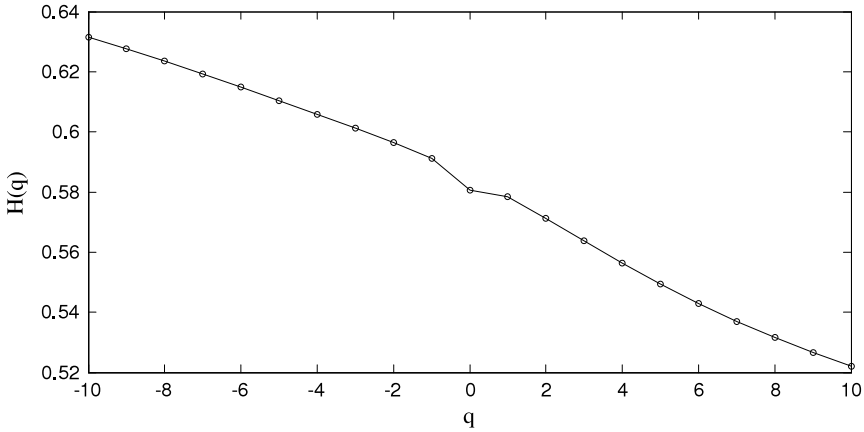

Fig. 1. The relationship of the $H(q)$ and q in the whole sample period.

It is well known that when $H(q)$ is constant for all q, the time series are mono-fractal. Otherwise, the series are multifractal. From Fig. 1, we find that different q can get the different $H(q)$, which means that, the value of $H(q)$ is not independent on q. In a word, the Chinese stock market has the multifractal feature, and the same conclusion also can be proved by Refs. [17,18,21].

4.2. Some forecasting results and MCS tests

The one-day ahead volatility forecasts for different models are constructed using the rolling forecasting methodology discussed in Section 3. To be clear, Figs. 2 and 3 show only one piece of the forecasting results ($n = 1, 2, \ldots, 364$, about 364 observations) in which the realized volatility measures as the forecasting benchmark are displayed as stars. Fig. 2 shows the forecasting results of HAR-RV, ARFIMA-RV, ARFIMA-RBV and ARFIMA-MFV models and Fig. 3 the forecasting results of HAR-Log(RV), ARFIMA-Log(RV), ARFIMA-Log(RBV) and ARFIMA-log(MFV) models.

To get the value of T_R, T_{SQ} and their MCS p-values, we choose the block-length, $d = 2$, and use 2000 times of bootstraps. For more details about the bootstrap methods in Ref. [34], we take the same significant level as Hansen et al. [34], in which the level of α is equal to 0.1. If the MCS p-value is less than 0.1, which means that the model should be deleted from the set of M_0. In other words, if the MCS p-value is less than 0.1, the forecasting performance of the model is worse than other models, otherwise, it is better.

Table 2 shows the results of the MCS test. The first column is the volatility models that are HAR-RV, ARFIMA-RV, ARFIMA-RBV, ARFIMA-MFV, ARFIMA-MV and their corresponding ARMA-type models. And the first row represents the value of the loss functions. Table 2 shows that the forecasting performance of the HAR-RV model is better than realized variance and multifractal volatility models. Surprisingly, only one model—the HAR-RV model passes the MCS test, and the MCS p-value is equal to 1. Therefore, we can conclude that the HAR-RV model is more accurate and sensitive to the market variability.

However, many researchers have found that RV and MFV show the feature of high peak and fat tail, which can be verified in Table 1 of this paper. It is well known that the logarithmic form of RV and MFV is near to normal distribution, and furthermore, Ref. [29] points out that the logarithmic form of RV and MFV could improve the estimation of the high-frequency

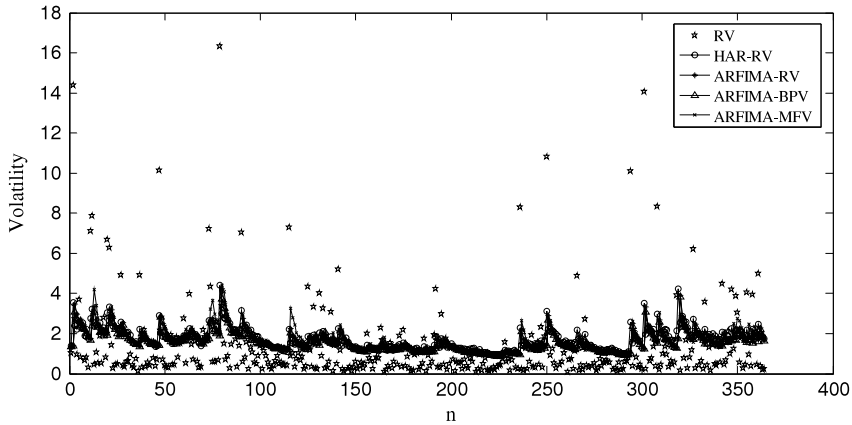

Fig. 2. One-day ahead volatility forecasts from HAR-RV, ARFIMA-RV, ARFIMA-RBV and ARFIMA-MFV, and realized volatility measures (as stars).

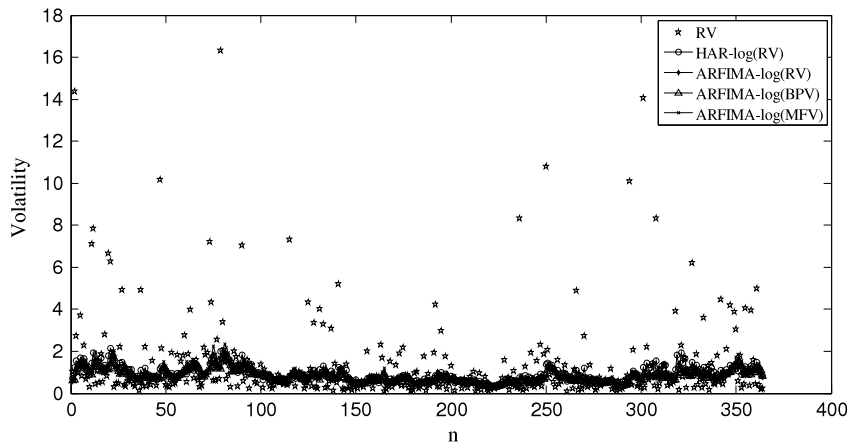

Fig. 3. One-day ahead volatility forecasts from HAR-Log(RV), ARFIMA-Log(RV), ARFIMA-Log(RBV) and ARFIMA-log(MFV), and realized volatility measures (as stars).

Table 2
The MCS p-value of the HAR-RV, ARFIMA-RV, ARFIMA-RBV, ARFIMA-MFV, ARFIMA-MV and corresponding ARMA-type models.

LOSS FUNCTION	*MSE*		*MAE*		*HMSE*		*HMAE*		*QLIKE*		R^2LOG	
	T_R	T_{SQ}	T_R	T_{SQ}	T_R	T_{SQ}	T_R	T_{SQ}	T_R	T_{SQ}	T_R	T_{SQ}
HAR-(RV)	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**
ARFIMA-RV	0.0001	0.0035	0.0000	0.0000	0.0027	0.0000	0.0000	0.0000	0.0001	0.0028	0.0000	0.0000
ARFIMA-RBV	0.0001	0.0035	0.0000	0.0000	0.0053	0.005	0.0012	0.0012	0.0001	0.0028	0.0000	0.0000
ARFIMA-MFV	0.0001	0.0035	0.0000	0.0000	0.0032	0.0034	0.0000	0.0000	0.0001	0.0028	0.0000	0.0000
ARFIMA-MV	0.0001	0.0035	0.0000	0.0000	0.0031	0.0000	0.0000	0.0000	0.0001	0.0028	0.0000	0.0000
ARMA-RV	0.0001	0.0027	0.0000	0.0000	0.0027	0.0000	0.0000	0.0000	0.0000	0.0014	0.0000	0.0000
ARMA-RBV	0.0001	0.0035	0.0000	0.0000	0.0027	0.0000	0.0000	0.0000	0.0000	0.0023	0.0000	0.0000
ARMA-MFV	0.0001	0.0035	0.0000	0.0000	0.0027	0.0000	0.0000	0.0000	0.0001	0.0028	0.0000	0.0000
ARMA-MV	0.0001	0.0035	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0028	0.0000	0.0000

volatility models and predicting performance. Table 3 shows that the HAR-log(RV) model has better forecasting performance than other models, such as ARFIMA-log(RV) and ARFIMA-log(MFV). Moreover, we also compare the forecasting ability of the HAR-RV and HAR-log(RV) model in MCS test. And the empirical results in Table 3 demonstrate that the HAR-log(RV) has better performance than HAR-RV at most loss functions. Furthermore, to affirm our results above more carefully in this paper, we also apply another method proposed by Diebold and Mariano [58], the DM test, and get the same results as the MCS test σ.

5. Conclusions

In this thesis, taking about 5-min high-frequency data of Shanghai Stock Exchange Composite Index (SSEC) from 1 January 2000 to 31 March 2013 as an example, we use the MCS test which is more robust and reliable than SPA to compare the

Table 3
The MCS p-value of the HAR-log(RV), ARFIMA-log(RV), ARFIMA-log(RBV), ARFIMA-log(MFV), ARFIMA-log(MV) and corresponding ARMA-type models.

LOSS FUNCTION	*MSE*		*MAE*		*HMSE*		*HMAE*		*QLIKE*		R^2LOG	
	T_R	T_{SQ}	T_R	T_{SQ}	T_R	T_{SQ}	T_R	T_{SQ}	T_R	T_{SQ}	T_R	T_{SQ}
HAR-log(RV)	**1.000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**
ARFIMA-log(RV)	0.000	0.0005	0.0000	0.0000	0.0009	0.0005	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
ARFIMA-log(RBV)	0.000	0.0005	0.0000	0.0000	0.0009	0.0005	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
ARFIMA-log(MFV)	0.000	0.0005	0.0000	0.0000	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
ARFIMA-log(MV)	0.000	0.0005	0.0000	0.0000	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
ARMA-log(RV)	0.000	0.0005	0.0000	0.0000	0.0008	0.0005	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
ARMA-log(RBV)	0.000	0.0005	0.0000	0.0000	0.0009	0.0007	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
ARMA-log(MFV)	0.000	0.0005	0.0000	0.0000	0.0001	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
ARMA-log(MV)	0.000	0.0005	0.0000	0.0000	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
The MCS test in HAR-RV and HAR-log(RV)												
HAR-RV	**1.000**	**1.0000**	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**1.0000**	**1.0000**	0.0000	0.0000
HAR-log(RV)	0.012	0.0123	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**0.5450**	**0.5450**	**1.0000**	**1.0000**

performance of several high-frequency volatility models in the out-of-sample volatility forecasting: HAR-RV, ARFIMA-RV, ARFIMA-RBV, ARFIMA-MFV and ARFIMA-MV. The empirical results of the Model Confidence Set (MCS) test indicate that the HAR-RV model outperforms other models at all loss functions. Surprisingly, only one model survived in the MCS test—the HAR-RV model. Furthermore, we also provide further consideration about above-mentioned models' logarithmic form, and the empirical results demonstrate that the forecasting ability of HAR-log (RV) model is better than other models. Finally, we also compare HAR-RV with HAR-log (RV) model in predicting future volatility based on MCS test, the empirical results tell us that HAR-log (RV) is the best model to model and forecast the Chinese stock market of the 22 high-frequency models in our paper. These conclusions may better direct the future research on derivative products pricing, portfolio allocation and financial risk management for investors and China's regulator.

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